

Question:

Tom's age is T years, which is also the sum of the ages of his three children. His age N years ago was twice the sum of their ages then. What is T/N ?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Reasoning + Answer:

<think>

1. Understanding the problem:

- Tom's age is T years, and it is the sum of the ages of his three children.
- N years ago, Tom's age was twice the sum of the ages of his three children at that time.

2. Setting up equations:

- Let the ages of the three children be a , b , and c .
- Therefore, $T = a + b + c$.

3. Years ago:

- N years ago, Tom's age was $T - N$.
- N years ago, the ages of the children were $a - N$, $b - N$, and $c - N$.
- The sum of their ages N years ago was $(a - N) + (b - N) + (c - N)$.

4. Given condition:

- According to the problem, $T - N = 2 \times ((a - N) + (b - N) + (c - N))$.

5. Simplifying the given condition:

- $T - N = 2 \times (a + b + c - 3N)$.
- Since $T = a + b + c$, we can substitute T with $a + b + c$.
- $a + b + c - N = 2 \times (a + b + c - 3N)$.
- $T - N = 2 \times (T - 3N)$.

6. Solving for T/N :

- We need to find T/N .
- From the equation $T - N = 2 \times (T - 3N)$, we can simplify it further.
- Divide both sides by N .
- $\frac{T}{N} = 3$.

<answer> 3 </answer>